

RDM & Statistics — Questions and Answers

Q1: Explain an RDM with a small example

Answer: An **RDM (Representational Dissimilarity Matrix)** is simply a table that tells you **how different every pair of stimuli is**.

Imagine we have **3 sounds**: dog, cat, car.

Suppose the brain response to each sound is measured using **3 neurons/voxels**:

Dog = [5, 4, 1]

Cat = [4, 5, 1]

Car = [1, 1, 5]

Dog and cat produce similar patterns, while car produces a very different pattern.

Step 1: Compare every pair.

We calculate a distance between the response patterns. For example, if we use correlation distance: **distance = $1 - r$** , where r is the Pearson correlation.

Imagine we obtain:

Dog $\leftrightarrow$ Cat = 0.1 $\rightarrow$ very similar

Dog $\leftrightarrow$ Car = 1.5 $\rightarrow$ very different

Cat $\leftrightarrow$ Car = 1.4 $\rightarrow$ very different

Step 2: Put those distances into a matrix.

Dog: [0, 0.1, 1.5]

Cat: [0.1, 0, 1.4]

Car: [1.5, 1.4, 0]

The diagonal is always 0 because we're comparing something with itself, and the matrix is symmetric.

The RDM tells us: "I don't care exactly how strongly each voxel responded. I care about which sounds produce similar or different patterns."

So here, dog $\approx$ cat, but dog $\neq$ car.

This becomes especially useful because you can make one RDM from **STG brain responses** and another from **a neural network's representations**, then compare the two RDMs. If they look similar, it suggests that **the model organizes the sounds in a similar way to the brain**.

Q2: Why did to use Pearson distance and Spearman correlation??

Answer: Because they are doing **two different jobs** in your RDM analysis.

1. Pearson $\rightarrow$ used inside each RDM

You start with the actual representations of each sound. You want to know: **How similar are the response patterns for sound A and sound B?**

So you calculate **Pearson correlation r** between their feature vectors and convert it into a distance: **$d = 1 - r$** .

For example: $r = 0.9 \rightarrow d = 1 - 0.9 = 0.1$, so the sounds are very similar.

You repeat this for every pair of sounds to **build the RDM**.

Pearson distance $\rightarrow$ constructs the RDM.

2. Spearman $\rightarrow$ used between two RDMs

Now imagine you've constructed a brain RDM and a model RDM. The question is different: **Does the model organize the sounds similarly to the brain?**

For this, you compare the corresponding RDM values using **Spearman correlation**. Spearman mainly cares about the **ranking** of the distances.

If both RDMs say A–B is the most similar pair, B–C is in the middle, and A–C is the most different pair, Spearman can be very high even if the exact numbers aren't identical.

Why not Pearson for both? Because when comparing brain and model RDMs, we often care more about whether they have the **same relative structure** than whether their distances have exactly the same numerical scale.

Pearson distance: “How different are these two sounds?” $\rightarrow$ build the RDM.

Spearman correlation: “Do these two RDMs tell the same story?” $\rightarrow$ compare RDMs.

Q3: Why are the trained and untrained runs treated as independent groups?

Answer: Because the **trained runs and untrained runs are not naturally paired with each other**.

Suppose you have trained models T1, T2, T3, T4, T5 and untrained models U1, U2, U3, U4, U5. Even though both groups contain 5 runs, there is no meaningful reason why $T1 \leftrightarrow U1$ should be a pair. T1 and U1 are separate model runs with their own random initialization/seeds.

That's why you treat them as **two independent samples** and ask whether the brain–model similarity scores are generally different between the trained and untrained populations of runs.

They would be paired if each trained model had a **directly corresponding untrained version**. For example, if you initialize model 1, measure it before training, then train that exact model and measure it again: $U1 \rightarrow T1$, $U2 \rightarrow T2$, $U3 \rightarrow T3$.

Then each observation has a meaningful partner, so you'd have **paired samples**.

In one sentence: **Trained and untrained runs were treated as independent samples because there was no one-to-one correspondence between individual runs in the two conditions.**

Q4: What do the error bars, Welch's t-test, and Holm correction tell us?

Answer: They answer **three different questions** about your trained vs untrained results.

1. Error bars → “How variable are the runs?”

The error bars show uncertainty or variability around the means. Small error bars mean the runs give relatively consistent results. Large error bars mean the results vary a lot between runs. Exactly what they represent depends on your code: they might be standard deviation (SD), standard error (SEM), or a confidence interval.

2. Welch's t-test → “Is the difference likely to be real?”

Seeing different means isn't enough, because the difference might arise from random variation between runs. Welch's t-test asks: **Is there evidence that the mean correlation is different between trained and untrained runs?**

The important output is the p-value. If you're using the conventional threshold $\alpha = 0.05$ and $p < 0.05$, the difference is statistically significant.

You use Welch's version because it **doesn't assume the two groups have equal variances**.

3. Holm correction → “Are we getting significance just because we tested many things?”

When you perform multiple statistical tests, the chance that at least one gives $p < 0.05$ just by chance increases. Holm correction makes the significance criterion stricter while accounting for the multiple comparisons.

Put together:

Error bars: How much do my runs vary?

Welch's t-test: Are trained and untrained means significantly different?

Holm correction: Does that difference remain significant after accounting for all the comparisons I made?

One important detail: **error bars overlapping does not automatically mean “not significant.”** The statistical test, not visual overlap alone, determines statistical significance.

Q5: Why does a p-value above 0.05 not prove that there is no difference?

Answer: Because a **p-value tells you about the strength of evidence**, not whether two groups are exactly the same.

Suppose you're comparing H_0 : trained and untrained have the same mean against H_A : their means are different.

If you get $p = 0.20$, you say: **We do not have enough evidence to reject H_0 .**

You cannot say: “We proved that trained and untrained are the same.”

Why? Imagine the true means actually are untrained = 0.20 and trained = 0.25. There is a difference of 0.05. But suppose you only have 5 runs and there is lots of variability. Your test might produce $p = 0.18$.

That doesn't mean the difference doesn't exist. It could mean **your experiment doesn't have enough**

statistical power to detect it.

So instead of writing “There was no difference between trained and untrained models,” it’s safer to write:
“We found no statistically significant difference between trained and untrained models.”

The key sentence to remember: **Absence of evidence is not necessarily evidence of absence.**